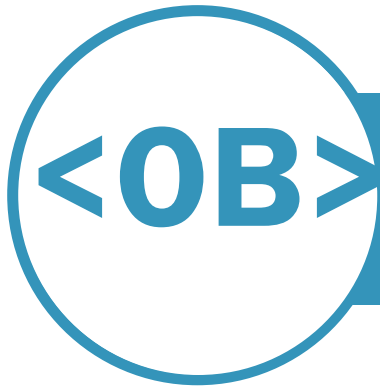




<OBENBET>

**<CS> | CAPSTONE PROJECT
HONOURS BACHELOR OF COMPUTER SCIENCE
(DATA ANALYTICS)**

STUDENT TEAM

Malik Henry, 4th Year Student
E: henrymal@sheridancollege.ca

Peter Mina, 4th Year Student
E: minap@sheridancollege.ca

SUPERVISOR

Prof. Paolo Treves
E: paolo.treves@sheridancollege.ca
Sheridan College

DOMAIN EXPERTS

Samuel Palubiski,
E: <https://www.linkedin.com/in/samuel-palubiski-275863201/>
T: 226-218-7726
Coordinator, Basketball Operations
at Canada Basketball

Simon McNeil,
E: <https://linkedin.com/in/mcneils/>

Senior iOS Developer, TD Bank

ABSTRACT

OpenBet is an online NBA betting analytics and social platform. It is designed to empower users in the rapidly expanding online sports betting sector. OpenBet's current focus is NBA basketball wagers. The global sports betting market is estimated to be valued at over \$100.9 Billion USD in 2024 [1]. This sector raises various ethical and operational concerns, including hidden odds setting practices, high house edges and predatory marketing strategies. OpenBet aims to provide users with tools to make more informed gambling decisions along with a social space to share gambling insights. OpenBet aggregates live odds from various sportsbooks for all NBA markets and compares against our own Random Forest Machine Learning model. Discrepancies between predictions compared to lines inform users of individual wager values. Arbitrage detection is provided to provide users with smart value opportunities that exploit market inefficiencies [2]. Our solution is not a betting site but a tool to promote responsible and informed decision making.

ABOUT CAPSTONE PROJECTS

TIMELINES • PROGRAM • SCHOOL

- **Sept 2025 – December 2025:** [Capstone Project Inception](#), 4-credit course (18 hours / week)
- **Jan 2026 – April 2026:** Capstone Project, 4-credit course (18 hours / week)

PROGRAM • SCHOOL

- [Hons. Bachelor of Computer Science \(Specialty in Data Analytics\)](#)
- [Applied Computing, Faculty of Applied Science and Technology](#)

Sheridan

STUDENT TEAM	1
SUPERVISOR	1
domain experts	1
ABSTRACT	1
ABOUT CAPSTONE projects.....	1
timelines • program • school	1
program • school.....	1
Introduction	3
Project Overview	4
Domain and Industry Overview	5
Problem Description.....	5
Solution Description	5
Data Analytics.....	6
Cloud Computing.....	6
Advanced Areas of Computer Science	6
Solution Impact	7
Solution Feasibility	7
Design and Construction	7
Deployment.....	7
Adoption.....	7
Project Requirements	8
System Context	9
Use-Cases	9
User interface	12
Project Architecture	26
Architecture Overview	27
System Components	29
Deployment Model	29
Project Plan	30
Iteration Plan.....	30

Risk Management Plan.....	31
Validation and Testing.....	32
Dataset and Model.....	32
• Testing Strategy.....	34
Validation Results.....	34
Cost Estimation & Feasibility Analysis.....	35
Conclusion.....	35
Project Suitability	36
Domain Expert Evaluation.....	36
User Testimonials	37
Alpha Release	37
Beta Release	37
Appendices	38
Bibliography.....	39

INTRODUCTION

This document acts as the project plan for OpenBet. It outlines the objectives, deliverables and development strategies our team will follow throughout the capstone project.

Sep 21, 2025	<u>Project Plan</u> : Initial Project Overview, Domain and Industry Overview, Solution description, Problem description, Solution Impact, Solution Feasibility
Oct 19, 2025	<u>Inception Release</u> : UI wireframes
Nov 16, 2025	<u>Architectural Release</u> : Project Requirements, Project Architecture
Dec 07, 2025	<u>Elaboration Release</u> : Validation and testing strategies that are being planned, conclusion section
Dec 09, 2025	<u>Defense Meeting</u> : Addressed Initial Feedback
Jan.14, 2026	<u>2nd Defense Meeting</u> : Further addressed additional feedback along with initial feedback
Jan, 26, 2026	<u>Project Elaboration Iteration</u> : Improving on feedback from 2 nd defence meeting and further development of the project
Feb 22, 2026	<u>Capstone Alpha Release</u> : Improving on Elaboration feedback and last development of functionality
Mar 23, 2026	<u>Capstone Beta Release</u> : Fine tuning of models, ensuring there is no bugs and a cohesive visual language
April 10, 2026	<u>Capstone Defence Presentation</u> : Presenting the capstone
April 17, 2026	<u>Capstone Final Release</u> : Using feedback from the capstone defence comitee

Table 1.1: Capstone Iteration/Release Schedule

PROJECT OVERVIEW

OpenBet is a capstone project developed by Peter Mina and Malik Henry, fourth-year students in the Honours Bachelor of Computer Science program (Data Analytics specialty) at Sheridan College. OpenBet is designed to be an educational web app for NBA basketball bettors that integrates real time NBA data with its proprietary Machine Learning algorithm to generate predictions to be contrasted with official sportsbook odds and lines. The current scope of the project focuses specifically on NBA basketball and includes the following:

- Real-time Analytics: Aggregating odds from multiple sportsbooks to display side-by-side with our internal ML predictions.
- Market Tools: Arbitrage Detection and Player Prop analysis.
- Social Community: A space for daily discussion threads for users to share predictions and insights.

OpenBet development receives support from industry professionals to ensure project viability and functionality. Samuel Palubiski, Coordinator of Basketball Operations at Canada Basketball, advises us on our projects basketball analytics and statistical relevance. Simon McNeil served as an expert on the software architecture of our project.

DOMAIN AND INDUSTRY OVERVIEW

We believe that OpenBet falls within the Gambling Industry sector (7132). This is the Gambling Industries group [3]. With a primary focus on the rapidly growing sports betting sub-sector. Ever since the passing of the single event sports betting Bill C-218 in 2021 the industry has aggressively shifted towards digital platforms. This has made sports betting and online sportsbook usage soar at unseen rates. The industry is primarily a digital economy now. The Canadian sports betting market alone was valued at over \$7 billion USD in revenue in 2024 [4]. Its growth is projected to grow at a compound annual growth rate of 12% through 2030. Many of these new adopters come from younger demographics individuals. Ontario alone saw a 72% year over year increase for 2023-24 gaming revenue [5]. This unparalleled growth highlights the need for tools like OpenBet to empower the public. Betting volume and accessibility has soared while user oriented education tools have not kept pace.

PROBLEM DESCRIPTION

The problem OpenBet seeks to solve: is that bettors are wagering while uninformed. While the sports betting industry has grown massively seeing over \$142 billion USD wagered in the US in 2024 [6]. The average user has no idea if the odds they are betting on are fair. These sportsbook platforms use complex algorithms to set lines that give the House a huge advantage and they aggressively market these bets to young or inexperienced bettors who are not able to tell the difference. A recent example of this happened in January 2026, when FanDuel Sportsbook was fined \$350,000 CAD by the Ontario regulators (AGCO) [7]. They were caught accepting suspicious bets that had "integrity concerns" without stopping them or warning the public. This proves that while the sportsbooks have the tools and data to see activity that average bettors do not have. The average bettor serves as the primary beneficiary of the OpenBet solution. Specifically the young adults who are most target by sports gambling advertising.

SOLUTION DESCRIPTION

OpenBet's solution is a secure, web app platform that makes betting on basketball much easier by making this “modern” sports betting transparent, data driven, and easy to understand. We will use APIs to offer accessible real time odds comparison/detection, social community and arbitrage detection, all delivered directly to users through an intuitive web interface. The full stack infrastructure should be fully cloud based, with scalable cloud databases and APIs to collect, process, and serve live odds and player statistics as well as using machine learning models for real time line setting and their possibility to lose, probability assessment and for arbitrage detection. Advanced areas of computer science are cloud computing, data analytics, and machine learning. Machine learning will be used to calculate dynamic odds and where a loss can often happen, to help players avoid lines set at “incorrect” values. Later in this report, we will go through how the machine learning models are trained and deployed, how leverage in cloud infrastructure secures data handling and app reliability, and how real time analytics and arbitrage logic makes OpenBet a tool anyone can use.

Data Analytics

OpenBet uses data analytics as the backbone and brings together a cloud backend, basketball data APIs, machine learning components, and the front end, giving bettors a clean, informative experience. All of the in game statistics, live odds, sportsbook feeds, user selections, and historical results are captured and processed by API(s) which will then be filtered to find any and all arbitrage opportunities, as well as lines that are more likely to lose. RESTful web services show this preprocessed data to the frontend, where users can view, filter, and compare outcomes. The architecture of the system separates raw data, data cleaning, and ML models into different services to allow for easy scalability. Data analytics is crucial in attacking these problems with high-risk sports betting: without automated feeding (of data from sportsbooks), cleaning, and analysis, it is very difficult to reliably find market mistakes, explain risk, or offer fair line setting. This is because all the big companies are in coercion together to keep lines the same across platforms since there is no incentive to offer “better” odds since the competition will make more money. These techniques also support responsible gambling by offering transparency, information on the betting market, as well as ways to make money based on their mistakes.

Cloud Computing

The entire OpenBet infrastructure is hosted in the cloud. This allows for scalability, real time performance allowing with efficient operation costs. The python backend is deployed on Render, a cloud platform that allows us to continuously deploy updates from GitHub. For our database we use Google Cloud Firestore’s NoSQL database. It features real time data syncing for when our users post on our platform’s discussion threads. Cloud computing helps provide users with the live modern experience they expect. The NoSQL database schema was specifically chosen for its document based structure as it aligns with our project’s social features. Features two collections for user and discussion threads. They are stored as JSON-like documents and allow for rapid querying to enable live synchronization across the platform.

While the current free tier hosting on render is suitable for the final release, the system is setup to be scaled larger if demand requires. Upon expansion of our API loads or increased heavy traffic our first step will be to upgrade to

Render's paid tiers. This provides us with a quick and efficient route to vertically scale our project. If OpenBet reaches a point of massive traffic that requires complex availability configurations, our backend infrastructure will transition to AWS. This would allow use to implement load balancing to distribute incoming traffic across multiple containerized backend instances. This would allow for improved user experience with maximum uptime.

Cyber Security

OpenBet incorporates various cybersecurity strategies within the application layer and overall system infrastructure. Google Oauth 2.0 via firebase authentication is used to handle user authentication and profile management. We have delegated the task of handling sensitive user data like password to a trusted enterprise-grade security provider. All communication between the client's browser, firebase frontend, render backend, firestore database is strictly enforced over HTTPS. This secures the data packets in transmission using TLS/SSL encryption. This neutralizes any threat of malicious actors eavesdropping or stepping in the middle of data transmissions. All environment variables are managed within GitHub Secrets, Render Environment Variables and AWS secret managers which are all backed by zero-trust compliance preventing any credential exposure. To secure our database and site from injection attacks the social discussion threads feature input validation and sanitization protocols. All submitted texts will be stripped and left as safe plain text.

Advanced Areas of Computer Science

Machine learning is the heart of the OpenBet platform. We use it to provide predictions for individual matches and player props. We built a random forest regressor model that is trained on historical NBA data to predict specific outcomes like player points or final team scores. Our research shows that random forest models perform well for sport predictions as they are well suited for the non-linear relationships present in NBA basketball [8]. Recent studies have even shown that random forest regression models can predict NBA player scoring output with high reliability [9]. By comparing our machine learning model's predicted score against the sportsbook's market line users are able to outline discrepancies as well as some insight into how the line has been set.

SOLUTION IMPACT

OpenBet looks to empower its user to make informed gambling decisions. Helping users better understand probabilities through value calculations and probability visualization will encourage users to approach their gambling from a stronger foundation of understanding. We aim to provide both educational and social impact amongst users of our platforms.

SOLUTION FEASIBILITY

Design and Construction

We believe our three-tier architecture system involving the React Frontend, python backend and API integration, ML model data layer to be sufficient for our projects goal. We intend to leverage publicly available APIs for our NBA data

along with our sportsbook data to our benefit. We plan to pair this with open source machine learning libraries we have learned throughout our time here at Sheridan. These resources provide us with accessible and low costs tools to build our project

Potential Risks: Limited Team Size, Time Constraints, Poor Model Performance

Deployment

We strongly believe a Prototype of OpenBet is feasible within the semester timeframe. We don't handle any sensitive data nor do we handle payment or money. These frees us up from any compliance or security requirements. The heart of the project of comparing ML model prediction to sportsbooks lines can easily be achieved. The main point of contention would be the scope of the predictions were able to deliver. Our highest goal would to be able to provide predictions for various sports. Our minimum goal is to deliver basketball coverage.

Adoption

We believe adoption for OpenBet to be practically feasible. We are an open free to access online tool. We appeal to sports enthusiasts, academic users and gamblers. This creates a sizeable user pool to try to attract. We have very few barriers stopping potential users from adopting. We do not require any subscription, fees or signups to use our products. The main risk that jeopardizes our adoption is our prediction accuracy. Poor model prediction would undermine the entire project's value.

PROJECT REQUIREMENTS

This section defines how OpenBet operates and interacts with the outside world. It outlines the core functions of the system such as real time odds tracking, arbitrage detection and social interaction.

SYSTEM CONTEXT

Figure 1.1 serves as the system context diagram. It identifies the boundaries of the system along with the actors that interact with it. Our system connects two primary actors, guest or authenticated users to the platforms there main functional areas. FA1 UI and Dashboard the presentation layer for odds and predictions, FA2 Predictive Model our machine learning model engine that generates the predictions and finally FA3 User Management and social our authentication and community interaction area.

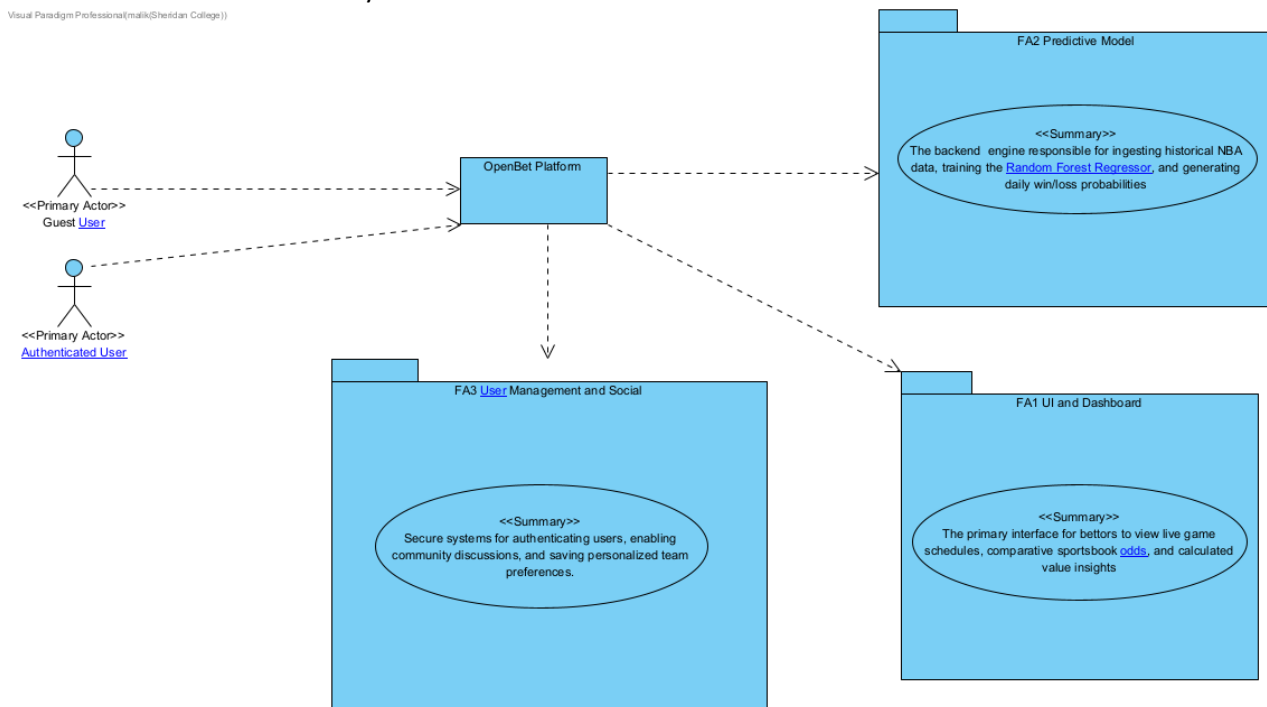


Figure 1.1: System Context Diagram

USE-CASES

Functional Area	Use-Case	Description
1 UI and Dashboard	View Games	Displays all basketball games and basic odds.
1	View Prop Bets	Displays player prop markets.
1	Arbitrage Opportunities	Users view risk-free profit opportunities where the system compares odds across sportsbooks
1	View Game odds and 2 predictions	Users compare official sportsbook lines against the OpenBet AI's predicted margin
2 Predictive Modeling	Ingest and Preprocess Data	The system ingests raw CSV data, normalizes Team IDs, and calculates rolling 10-game averages
2	Train Random Forest Model	Admin executes the offline training pipeline to update the .pkl model artifact
2	Generate Daily Predictions	An automated Github Action triggers the prediction pipeline daily to generate the todays_data.json file.
3 User Management and Social	Authenticate via Google Login	Users sign in using OAuth 2.0 to access restricted features
3	Post Comment to Thread	Authenticated users post text-based comments to the daily discussion board
3	Manage Favorite Teams	Users "Star" specific teams to prioritize them on their dashboard

Table 2.1: Functional Areas and Use Cases (cont'd)

Each functional area has been visualized with a use case overview diagram. Figure 1.2-1.4 depicts in order: FA1 UI and Dashboard Use Case Diagram, FA2 Predictive Model Use Case Diagram and FA3 User Management and Social Use Case Diagram.

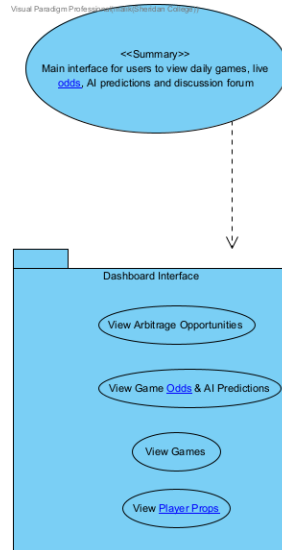


Figure 1.2: FA1 UI and Dashboard Use Case Diagram

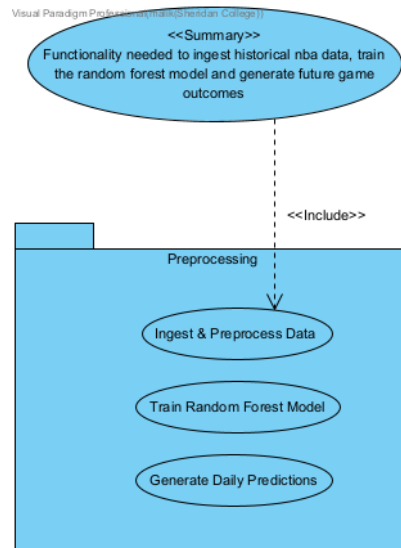


Figure 1.3: FA2 Predictive Model Use Case Diagram

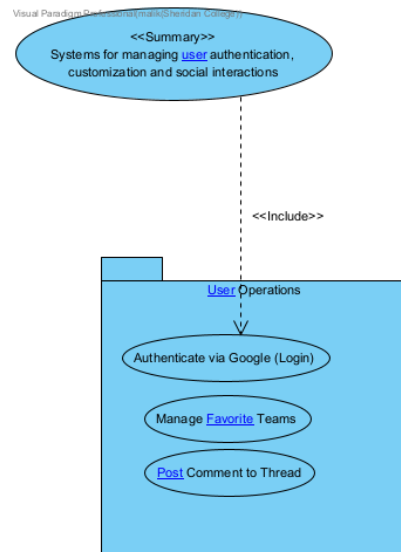


Figure 1.4: FA3 User Management and Social Use Case Diagram

USER INTERFACE

The user interface requirements define how users interact with the OpenBet system.

During the Inception Release, UI mockups are represented as wireframes; future releases will include working interfaces with real data.

The wire frames below represent the blueprint of the application. Focusing more on layout and navigation flow.

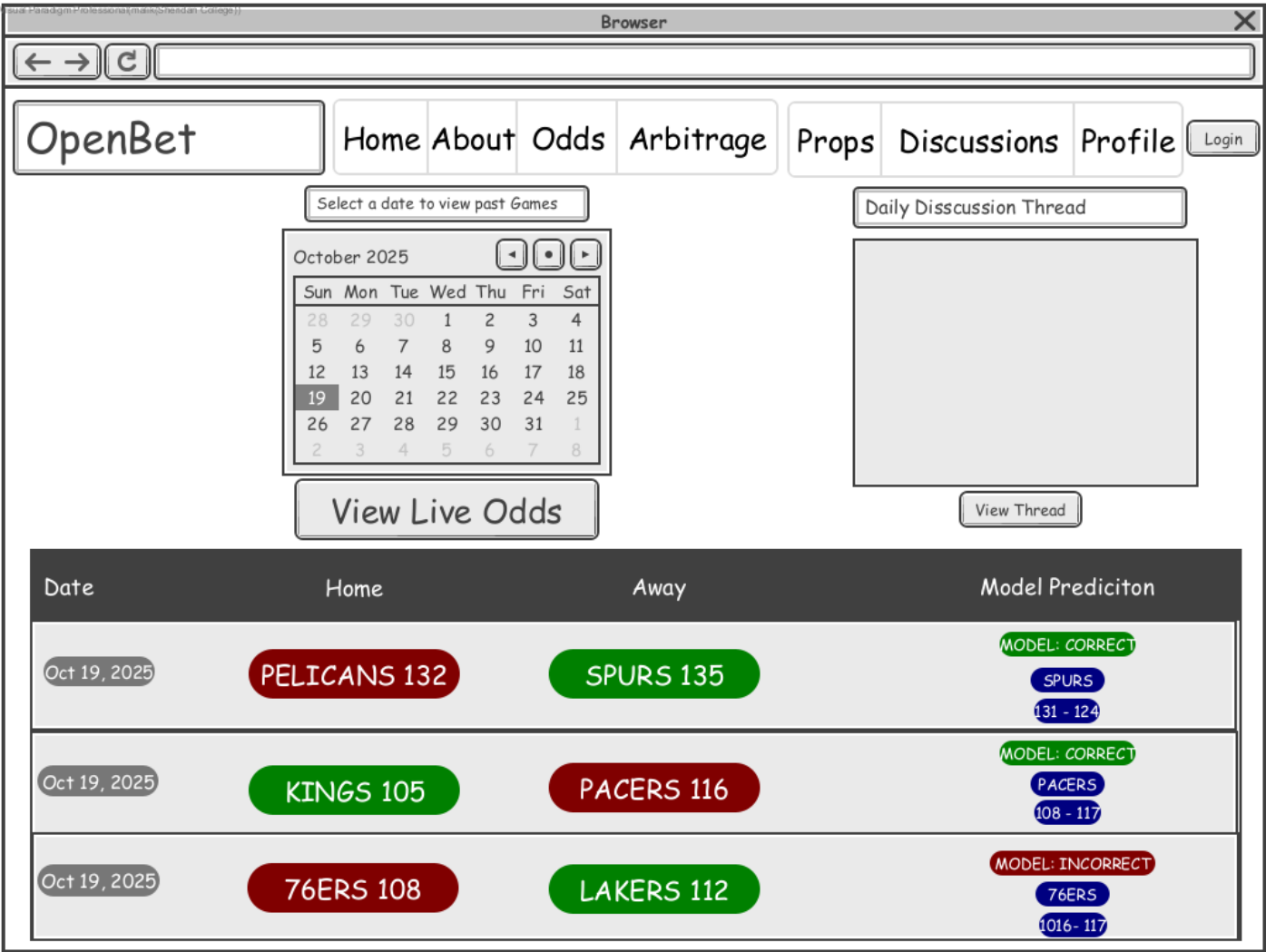


Figure 2.1: Home Page

Initial wireframe of the main page. Layout includes date picker, nav bar, game scores and discussion thread.

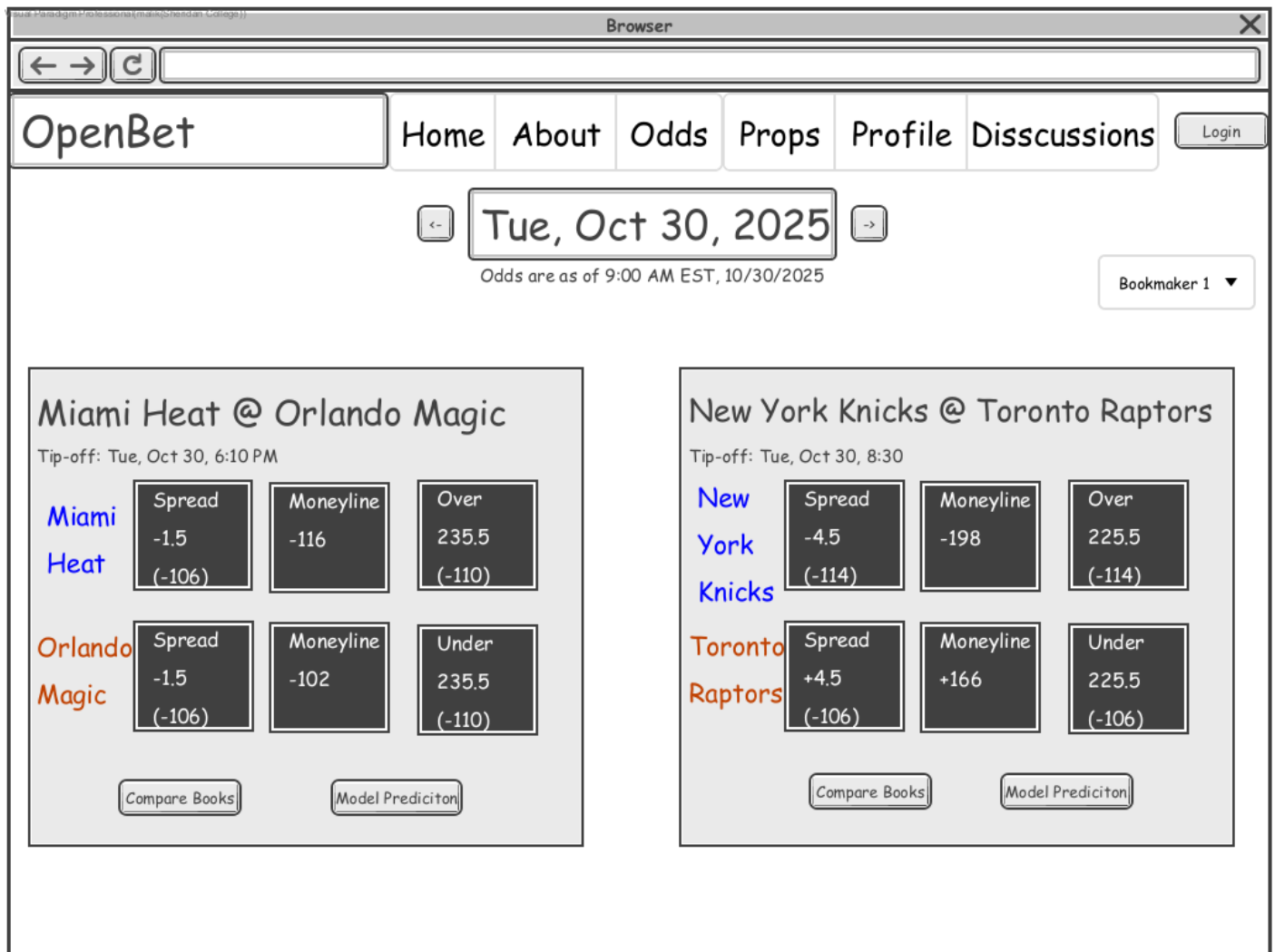


Figure 2.2: Odds Page

Initial layout for odds card page. Displays all odds for games on selected date.

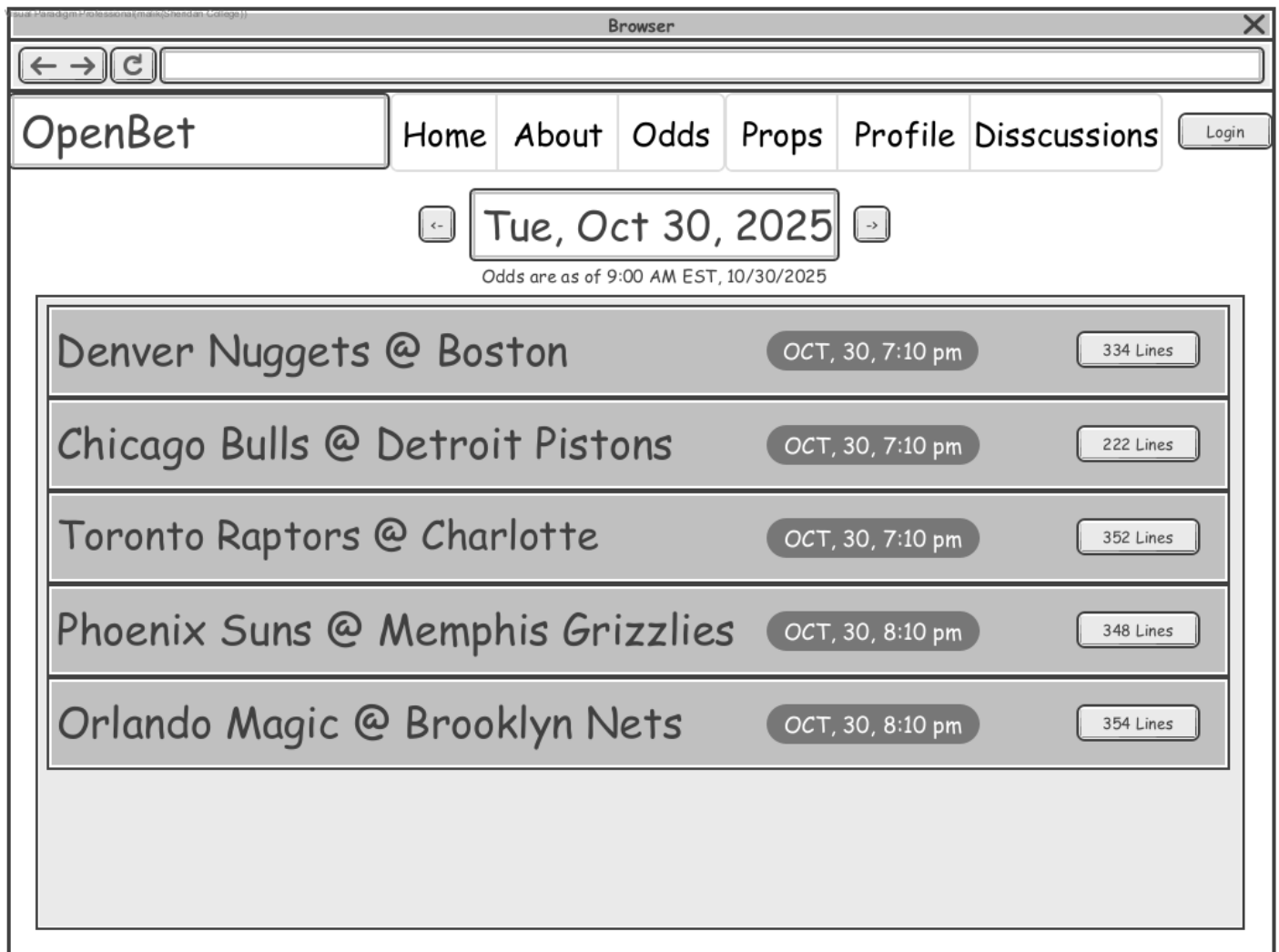


Figure 2.3: Player Props Page

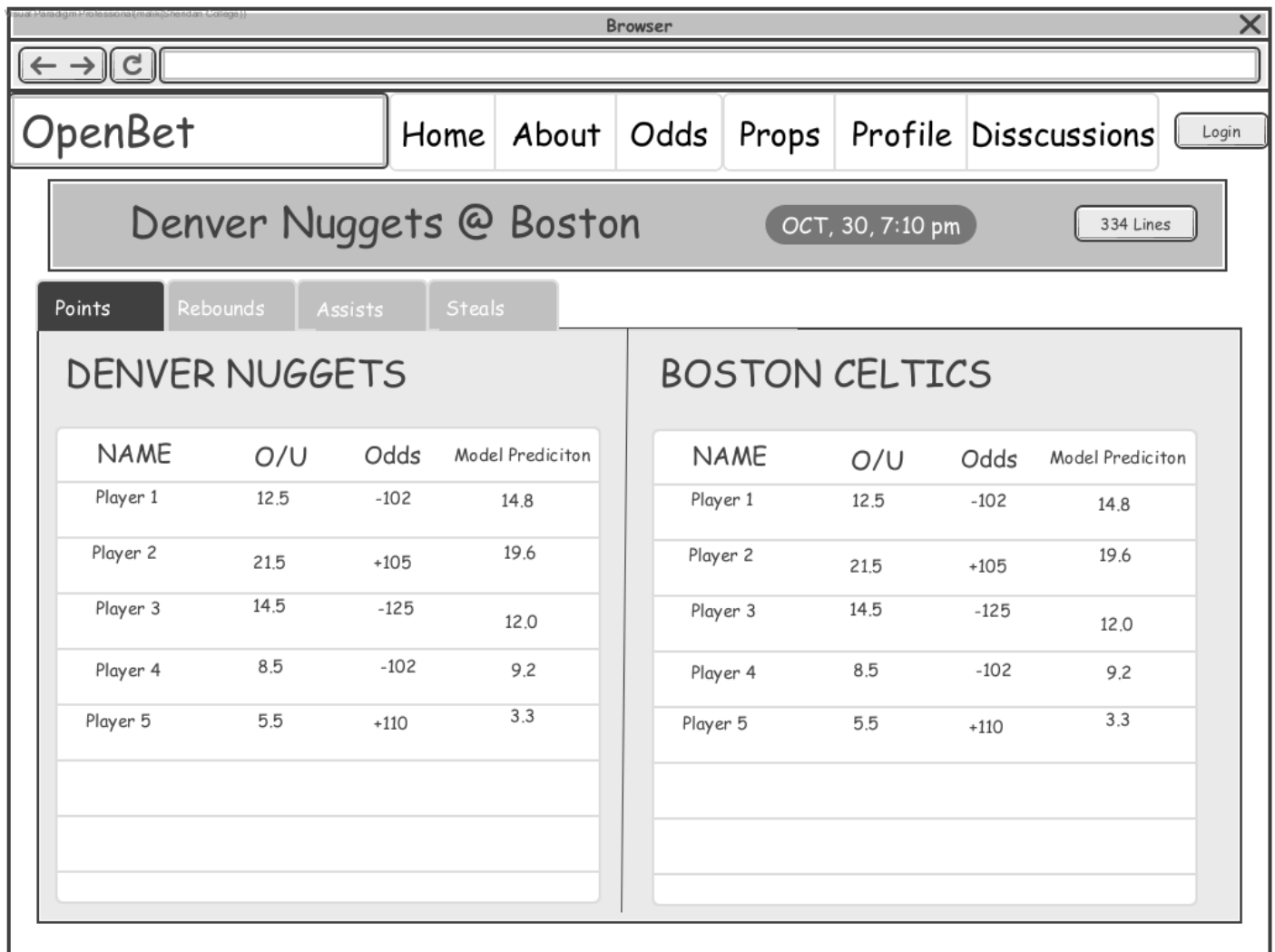


Figure 2.4: Player Props page expanded

Figures 2.3 and 2.4 are the player prop page layouts. Displays all markets for individual players available for the selected game.

The screenshot shows a web browser window titled "Browser" with a tab for "OpenBet". The browser's address bar is empty. The OpenBet website has a navigation bar with links: Home, About, Odds, Profile, Props, Discussions, Arbitrage, and a "Logo..." button. Below the navigation bar, there are two tabs: "Profile Details" (selected) and "Comment History". The main content area displays a user profile for "@Username". To the left of the username is a placeholder for a profile picture, represented by a square with an 'X' inside. Below the username, there are three input fields: "First Name" with the value "John", "Last Name" with the value "Appleseed", and "Favorite Team" with a dropdown menu showing a dark grey box and a downward arrow. Below these fields is an "Email" field with the value "Johnappleseed@mail.com". At the bottom of the profile section is a large "Update" button.

Figure 2.5: Profile Page

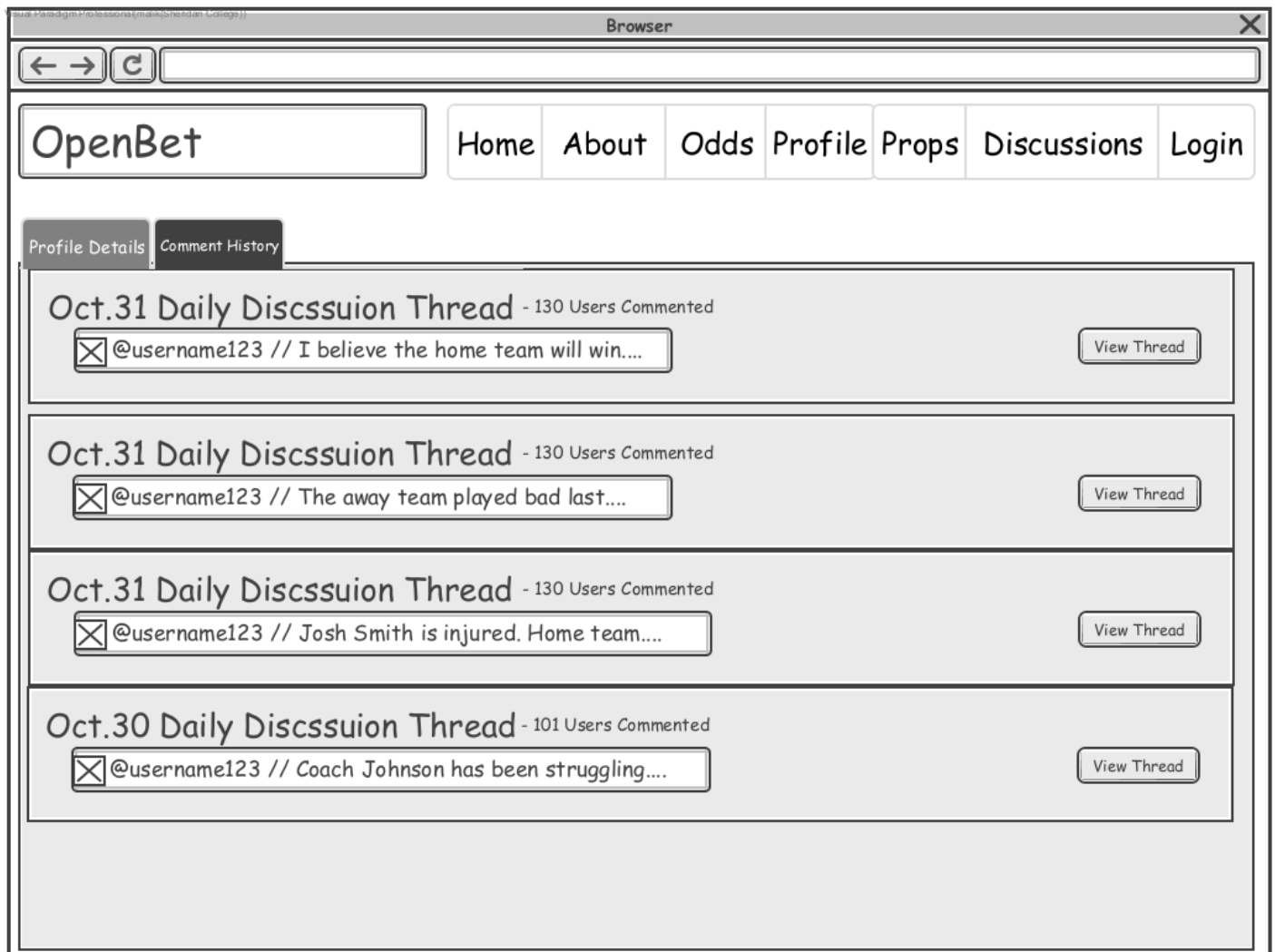


Figure 2.6: Profile Page expanded

Figures 2.5 and 2.6 are the profile page layouts. Users view all relevant profile information as well as their social posting history.

Browser

OpenBet Home About Odds Props Profile Discussions Login

DAILY DISCUSSION THREAD

< Tue, Oct 30, 2025 >

@username123

I dont beleive the raptors can win tonight. They are missing key palyers to injury and have been struiggling playing on the road

Reply

@username123

I love the hornets tonight. I believe their win streak makes it 10 games

Reply

Type here

Post

Figure 2.7: Discussion Page

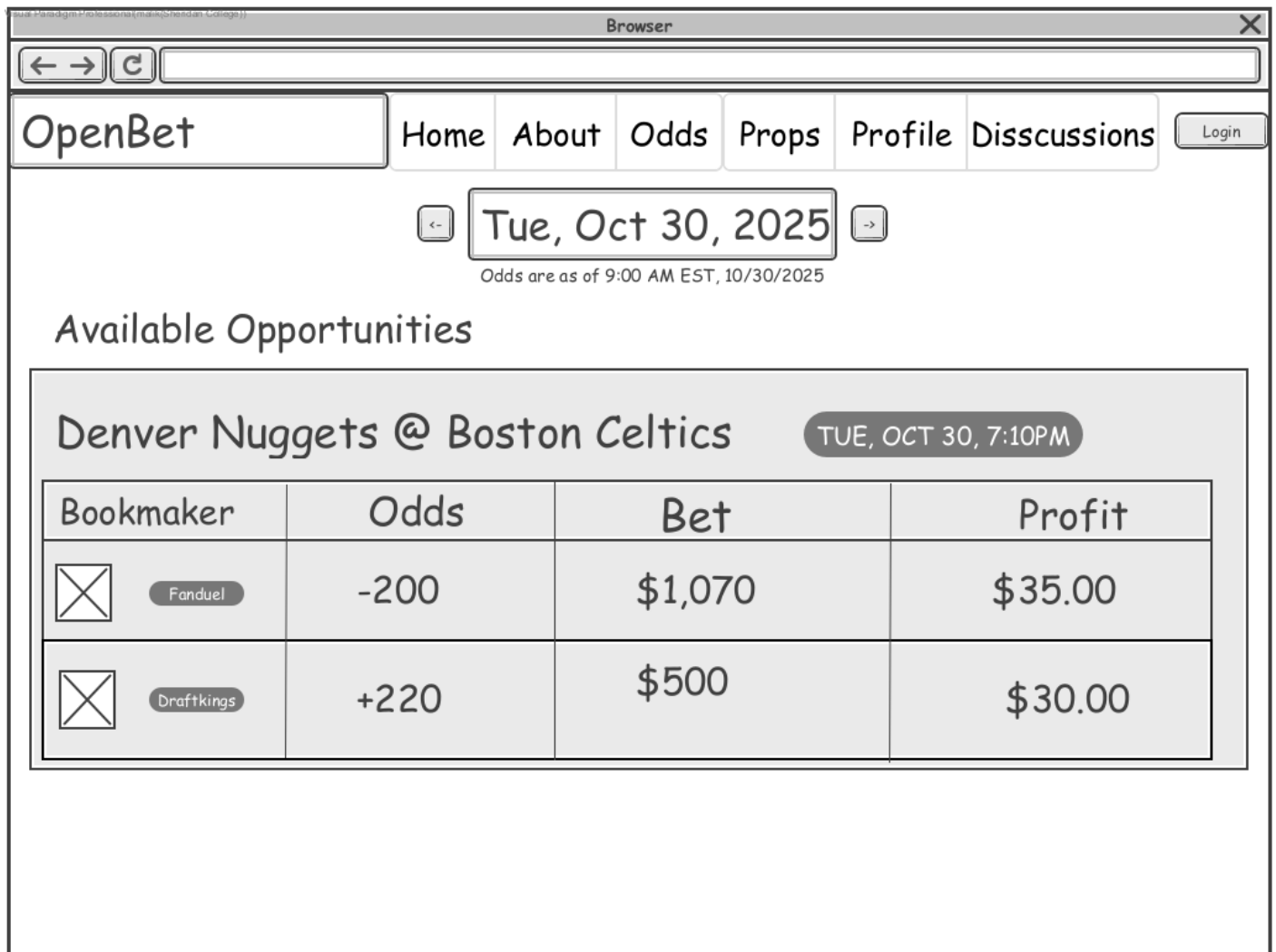


Figure 2.8: Arbitrage page

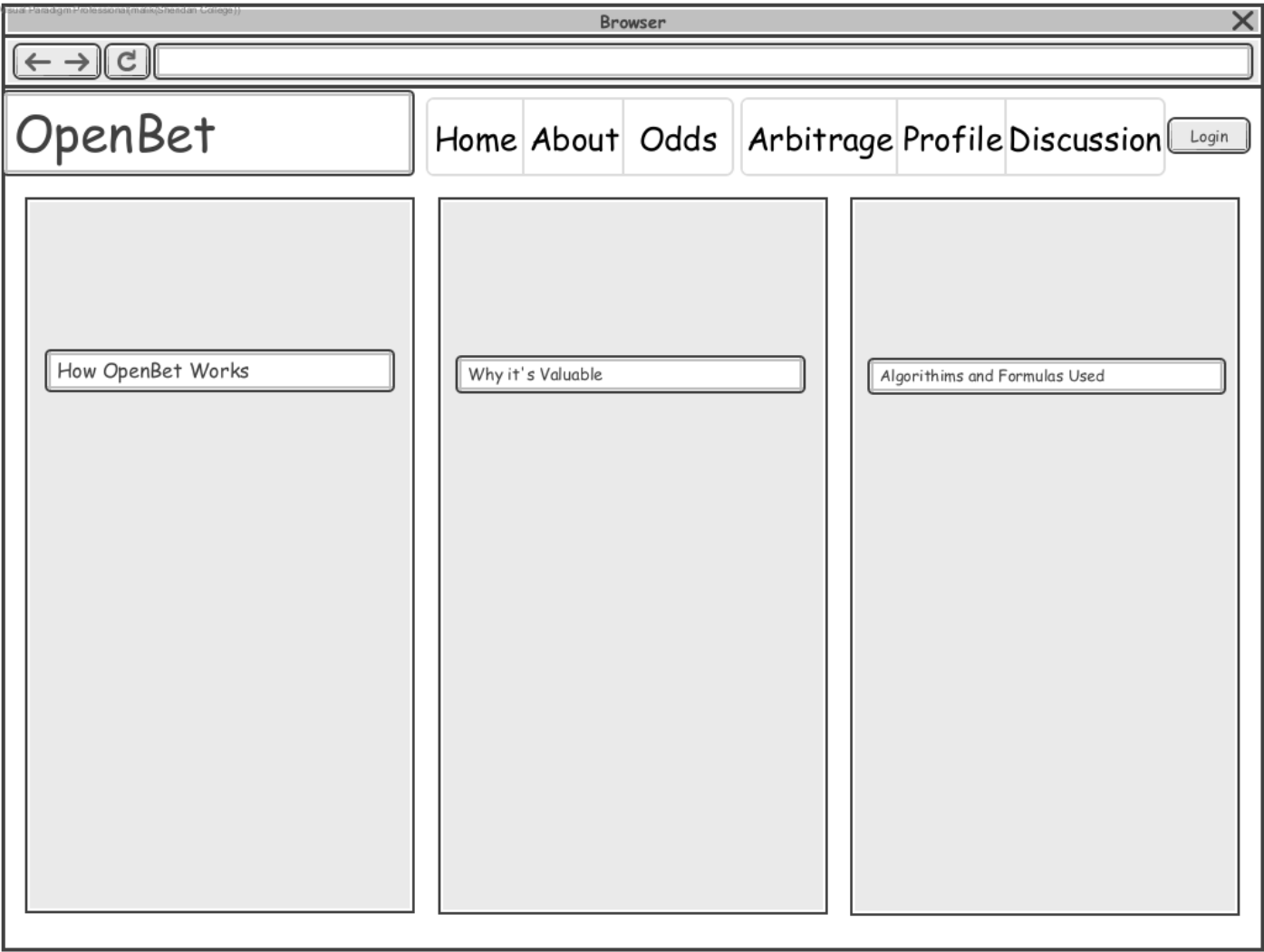


Figure 2.9: About Page

Current Wireframe Iterations

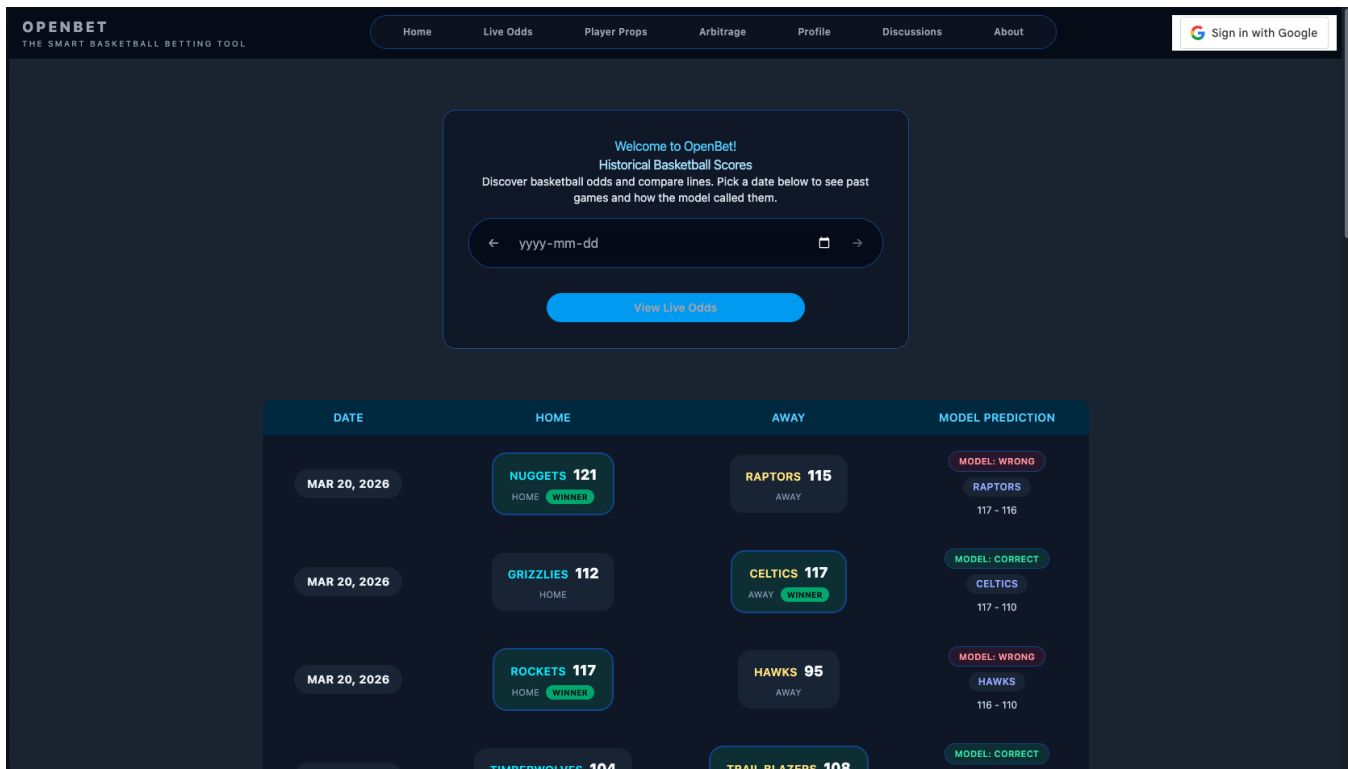


Figure 3.1: Current Home page

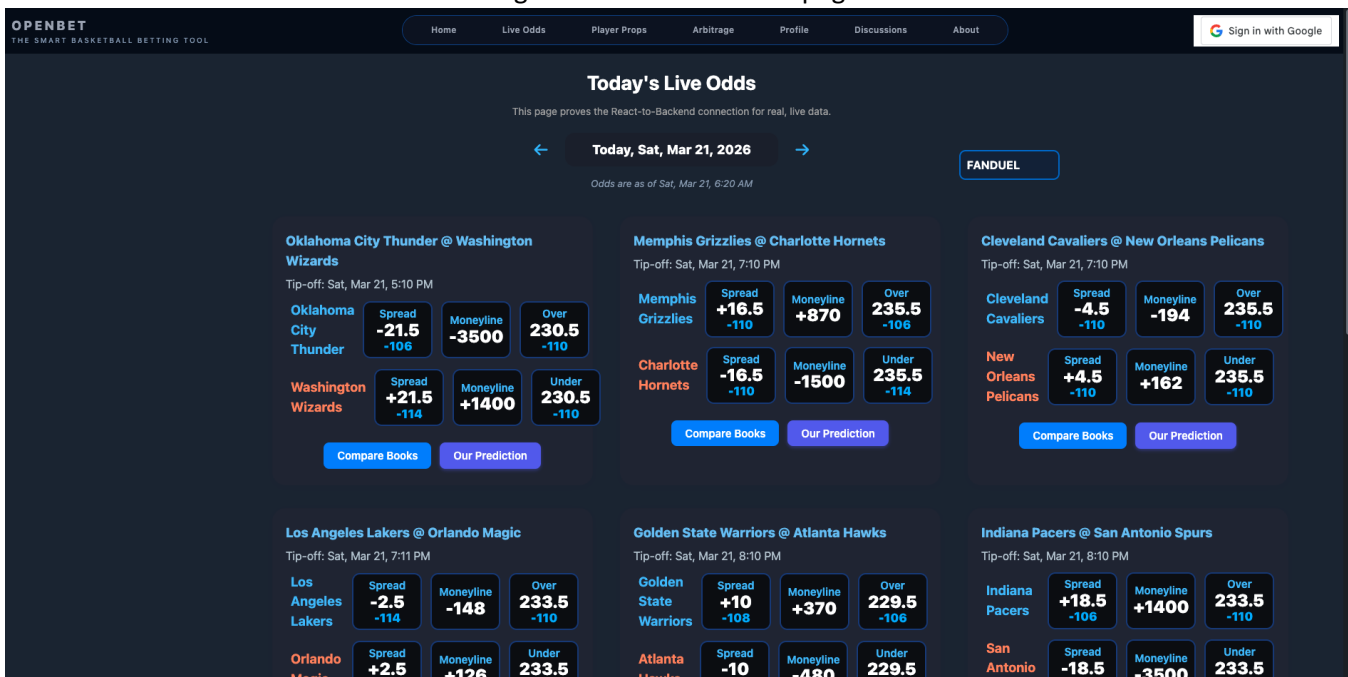


Figure 3.2: Current Odds page

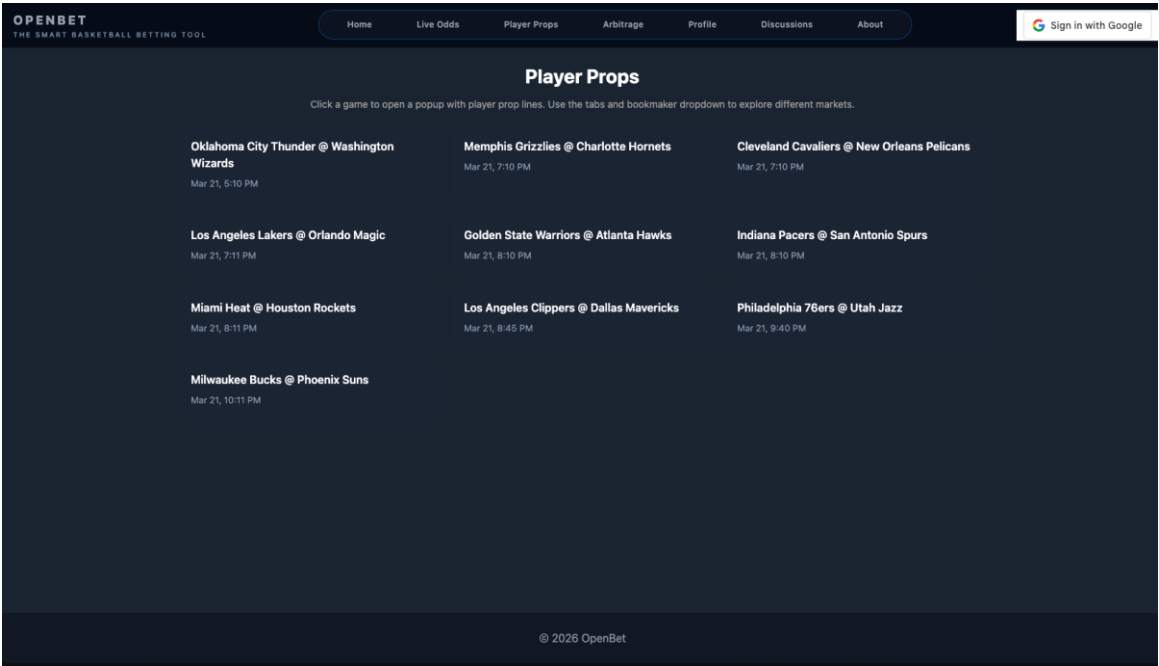


Figure 3.3: Current Player Props Page

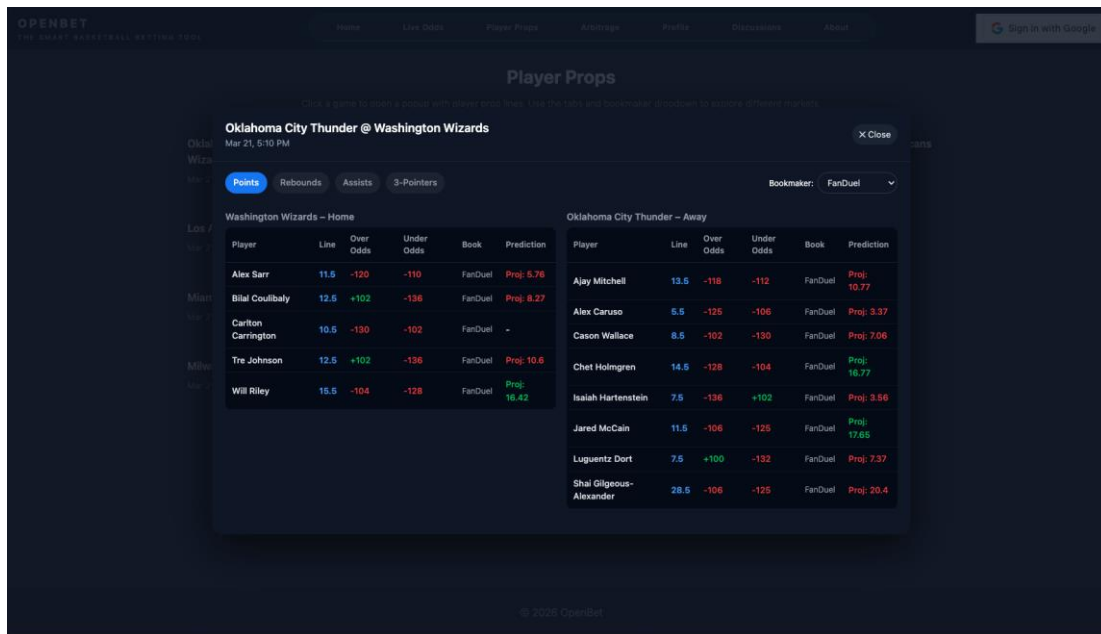


Figure 3.3: Current Player Props Page

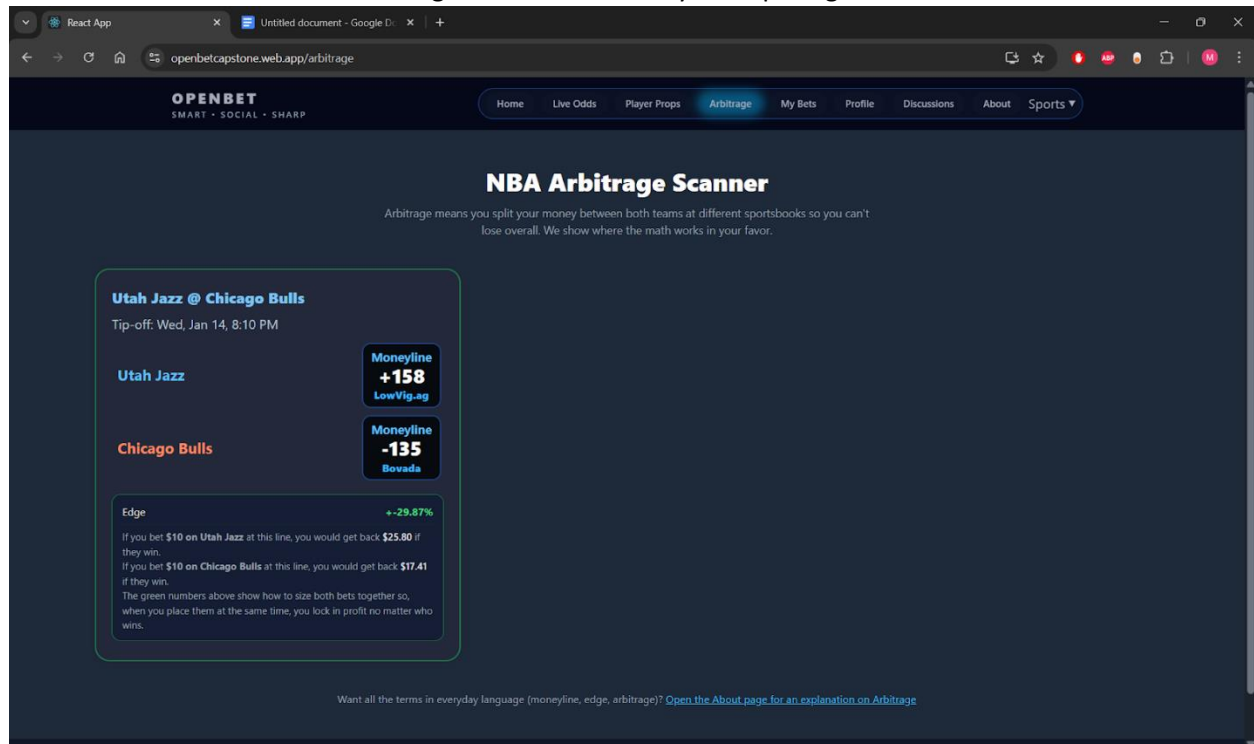


Figure 3.4: Current Arbitrage Page

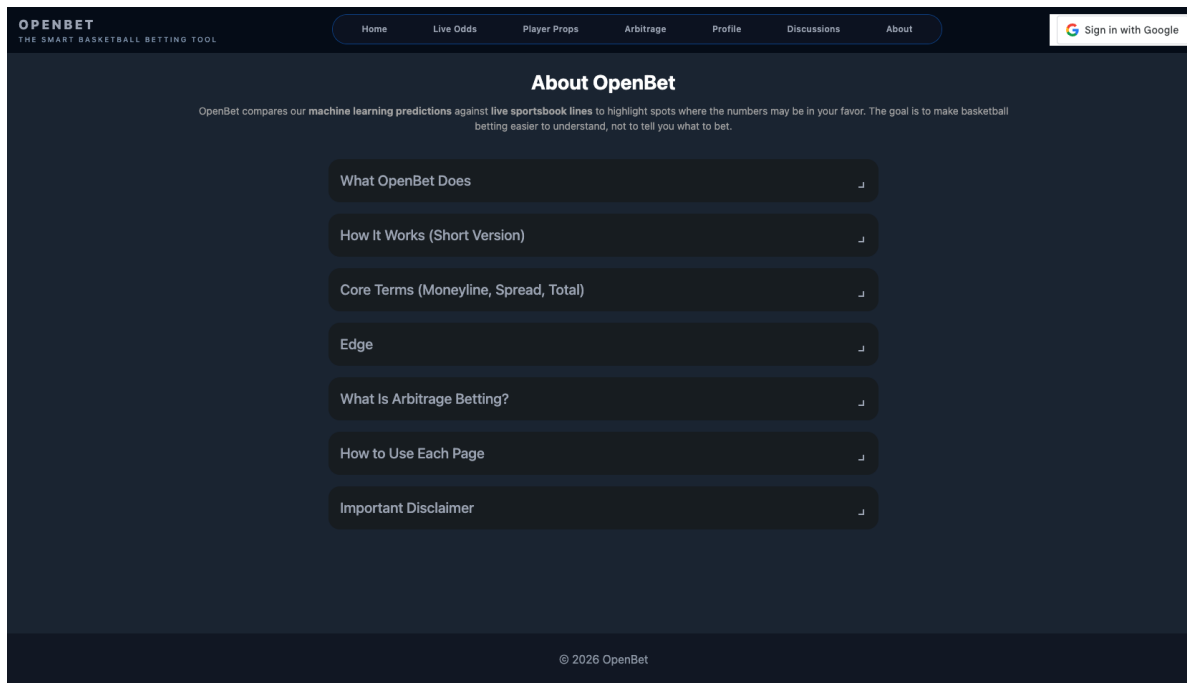


Figure 3.5: Current About Page

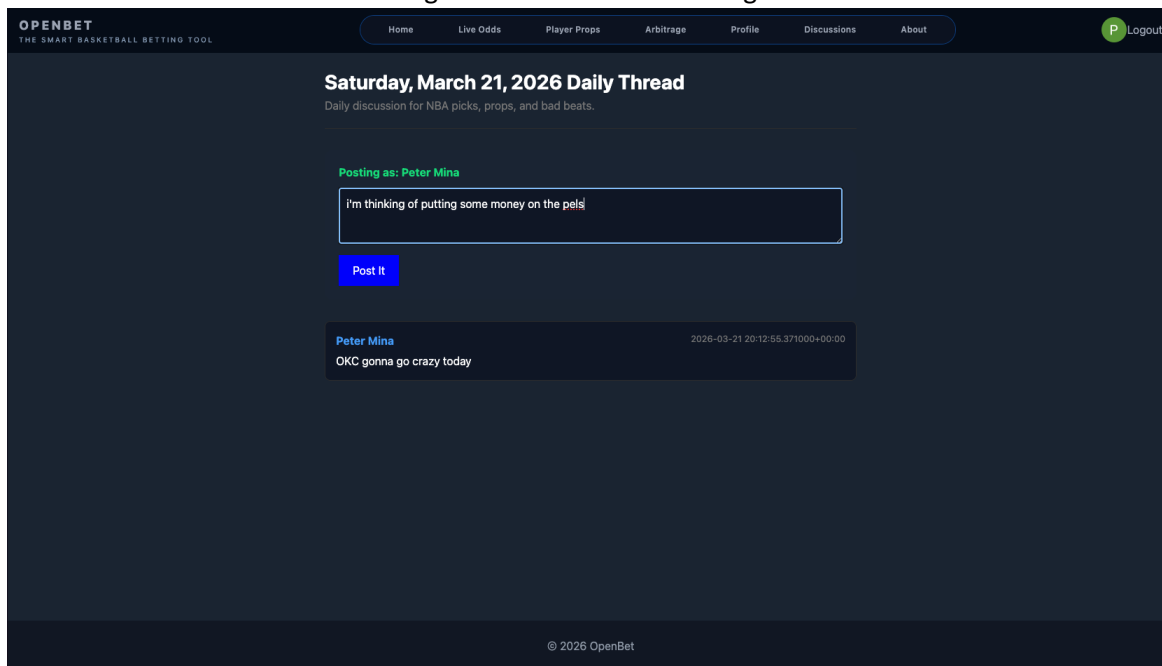


Figure 3.6: Current Discussions Page

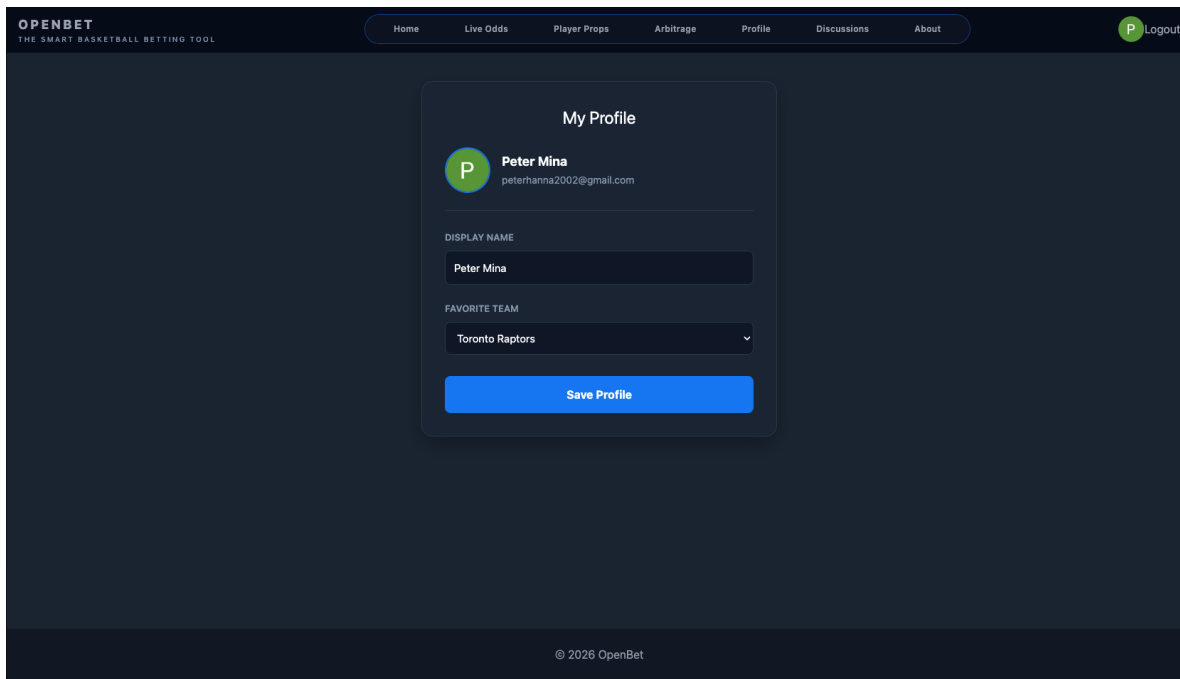


Figure 3.7: Current Profile Page

PROJECT ARCHITECTURE

This section contains a complete description of the OpenBet structure. This includes how its major components interact and how the chosen architecture supports the functionality of the system. Full technical implementation details can be viewed in the Design Model later in this section.

ARCHITECTURE OVERVIEW

The system follows a Layered Architecture separating things into Controllers, Business Services, and Data Objects.

UI Layer:

Our client side application is built with react. It is responsible for rendering the visual interface. It is comprised of 2 components. The View component which is the actual pages users see like the home page and odds cards. The second components are the client services used, google login authentication and the API calls.

App Layer:

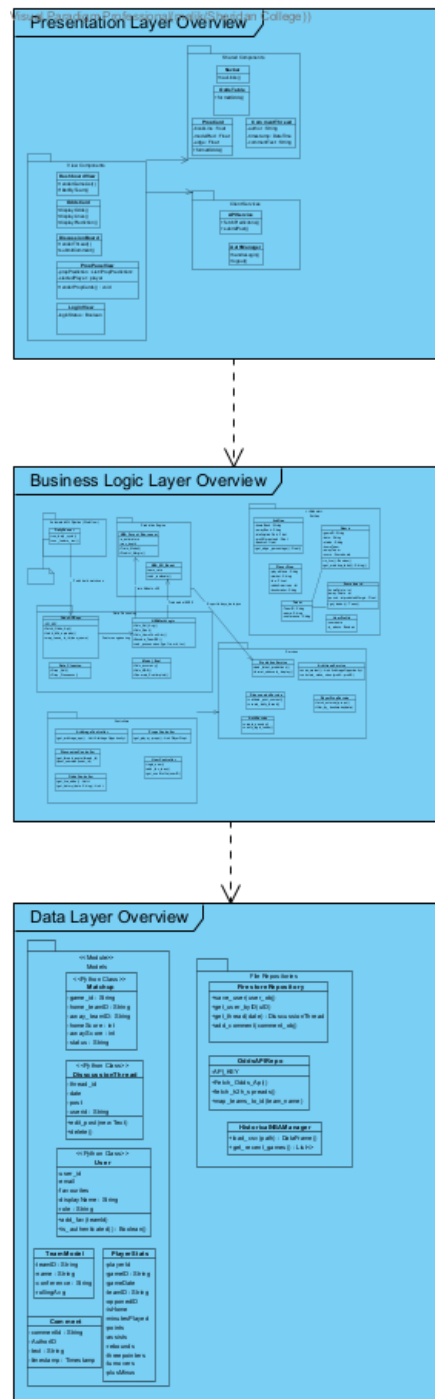
This acts as the server side of the project. It is built with Flask. It can be broken down into two components, Controllers and services. Controllers operate the core functions of the platform such as arbitrage, game odds, player props and discussions. The services encapsulate the logic of those functions.

Data Layer:

We use schemas to outline the structure of our stored data objects. Users are stored in NoSQL google firestore and game odds and predictions are stored in JSON files. The app layer serves it user request while the machine learning layer updates it with predictions.

Machine Learning Layer:

Independent layer responsible for the training and batch processing predictions. It contains the python scripts that ingest historical data and train the random forest model. It updates the data layer with daily JSON files containing the days new data which is to be consumed by the app layer.



SYSTEM COMPONENTS

This section details the primary components that make up the OpenBet platform. The system consists of the following Six components:

Client Component (React App):

The visual presentation layer that is running in the user's browser. It renders the user interface, manages the local session state login and executes fetch calls to the backend.

Web Hosting Component (Firebase):

This is our static asset server. It stores and provides our compiled frontend items like index.html and bundle.js. The separation of the two allows us to make quick changes to frontend services and displays without having to touch our backend server component.

Backend Server Component (Render):

This is our application logic provider. It is our Flask/Python environment that hosts our controllers and business services. Python scripts are executed that generate the daily data that gets sent back to the client.

Data Store Component (Google Firestore NoSQL):

The NoSQL database used to store user profiles and discussion threads.

External Data Provider (The Odds API):

This is our third party data provider. They supply us with the raw and real time data of sports betting line from all the major sportsbooks.

Machine Learning and Automation (GitHub Actions):

Our machine learning pipeline is executed via an automated GitHub actions runner Push/Deploy production setup,

DEPLOYMENT MODEL

The deployment model displayed below outlines the deployment environment of the OpenBet platform.

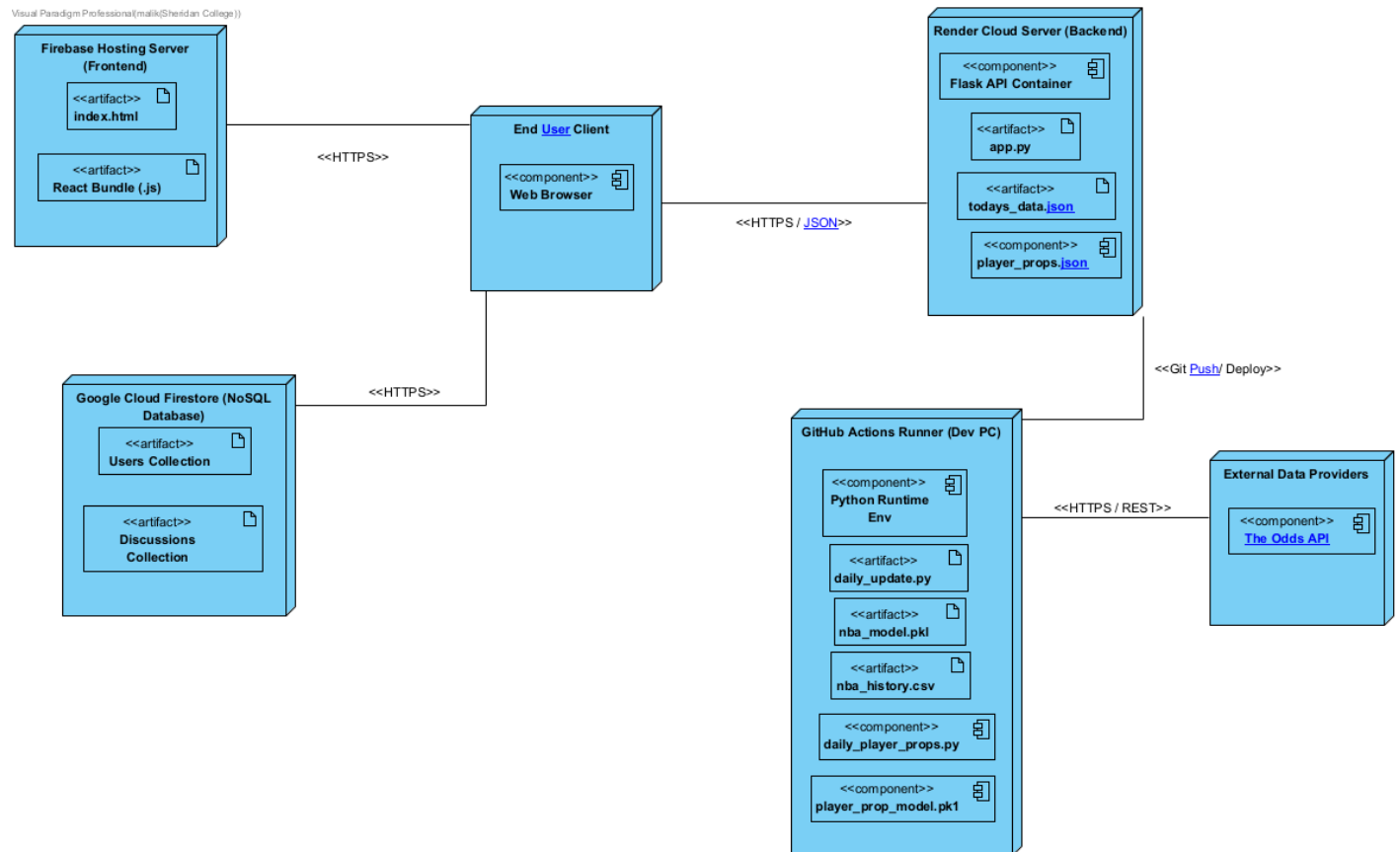


Figure 4.2: Deployment Model

PROJECT PLAN

This section outlines the project management strategy used to complete the OpenBet Capstone Project (Inception Phase). The team used an Agile Scrum approach. We used iterations to focus on incremental additions and improvements to the project.

Jira Project Management Plan: <https://sheridancollege-team-xwfxaw4f.atlassian.net/jira/software/projects/PMP/boards/200/timeline>

Jira Risk Management Plan: <https://sheridancollege-team-xwfxaw4f.atlassian.net/jira/software/projects/RMP/boards/201>

ITERATION PLAN

The construction of OpenBet followed a three-phase iteration plan. Each phase spans a month's length.

Iteration 1: Inception Release (October)

Scope Definition: We have finalized the core features of game, player predictions and arbitrage detection

Dataset Research: Our initial dataset was selected. It was limited in terms of historical accuracy. This dataset did not have the current NBA season's statistics. It was selected over alternatives as it was free in cost.

Development Environment Setup: The initial react frontend was set up along with the projects GitHub Repository.

Iteration 2: Architectural Release (November)

System Architecture: We defined our initial project structure with wireframes and class diagrams

API Implementation: We built the Python/Flask backend to successfully fetch and display game odds from our external Odds API data source

Iteration 3: Elaboration Release (December-January)

Dataset Update: We found a new dataset that is updated daily with current NBA data daily. We adjusted our preprocessing pipeline to account for a new dataset

Model Construction: We built our random forest regressor model to predict an NBA games outcome

Feature Addition: We added Arbitrage and User management features to the project plan

Iteration 4: Alpha Release (January-February)

Player Prop Page was fully implemented

Player Prop model was researched, developed, tested and deployed

Arbitrage Page logic was fixed

User Management was implemented (using Google Firestore)

Iteration 5: Beta Release (February-March)

Further research and development into the models (improvement was only seen in the Game Prediction Model)

Application/website is now fully automated

Fixed Visual language and some bugs

RISK MANAGEMENT PLAN

The risk management plan for OpenBet was made to be used alongside our Agile development cycle. We followed a continuous risk monitoring strategy where potential risks and threats to our projects development were identified,

categorized and mitigated during our iterative release plan. We used a Jira Risk Board to log all identified risks. Each risk was assigned a score based on severity and likelihood. This allowed us to prioritize the critical issues to address and monitor. Below are the major risks that were identified in our project's development cycle and how we managed them.

1. Dataset Quality and Availability (High Impact, Ongoing)
 - a. Risk: In the early iterations, the initial dataset we selected was lacking in recent data. This limited our models' effectiveness.
 - i. Mitigation: We pivoted to a new dataset that provided much more current data. We are still monitoring as this dataset may no longer be updated daily or it may become locked behind a paywall.
2. Scope Creep vs. Team Size (Medium Impact, Ongoing)
 - a. Risk: As a two-person team, we initially aimed to support multiple sports (NBA, NHL, NFL). We realized early on in development our current knowledge and expertise are not enough to support a project of that size.
 - i. Mitigation: We formally reduced the scope of the project. The NHL and NFL support was scrapped to focus exclusively on NBA Basketball. This allowed us to deepen the quality of the basketball features (adding Player Props and Arbitrage)
3. Model Accuracy and Complexity (Very High, Ongoing)
 - a. Risk: Our project relies on accurate predictions to provide any value to its users.
 - i. Mitigation: We adopted an iterative training approach. Instead of trying to build a perfect model immediately, we started with building our baseline model. Over the next semester, we plan to increase the features of the model to improve the model's performance over time.

Risk Management Jira: <https://sheridancollege-team-xwfxaw4f.atlassian.net/jira/software/projects/RMP/boards/201>

VALIDATION AND TESTING

This section will cover the validation and test suites set to verify the functionality of the OpenBet system. It includes the following:

Jira Link: <https://sheridancollege-team-xwfxaw4f.atlassian.net/jira/software/projects/OTPP/boards/234>

VP Link: [OpenBet2025.vpp://modelelement/nPdT31mGAqDIWA89](https://openbet2025.vpp://modelelement/nPdT31mGAqDIWA89)

DATASET AND MODEL

Dataset Evaluation

The OpenBet prediction engine is built off its Historical NBA Dataset sourced from Kaggle [10]. It is comprised of multiple dense csv files. Our current iteration operates using the Games.csv and TeamStatistics.csv. Our preprocessing of the data consists of event filtering to drop all non-relevant games. This is done using queries to drop all rows tagged

as “preseason” or “Allstar Game” under the gameLabel column. We addressed inconsistencies where gameId and teamId fields were mixed types. A custom clean_id() function was made to standardize all IDs into string format. This dataset was also chosen for its advanced metric columns. TeamStatistics.csv contains 48 columns with advanced metrics like pointsInThePaint, pointsFastBreak and biggestScoringRun. This fine granularity will allow for further model refinement in future iterations.

Model Selection and Training

For the Beta Release, the model was upgraded from a Random Forest Regressor to a Gradient Boosting Regressor (GBR). The Random Forest built multiple trees independently, but the GBR builds trees sequentially, so every new tree fixes the errors of the last one, this builds something that gets smarter as it goes on, which is perfect for the NBA.

This is supported by Ouyang et al. (2025) [11] who reviewed AI techniques in professional basketball and found that Gradient Boosting methods were consistently better than Random Forest, they attributed this to its amazing ability to model non-linear and interdependent nature of basketball stats, XGBoost was better at predicting outcomes across all five performance metrics.

We also tried to improve on the Player Prop model, but after researching and trying different models, there was no real improvement made, so we have elected to keep it the same.

Model Construction

Here we will outline the building and training of our GBR which is responsible for predicting the final teams score. We found raw statistics were insufficient for predicting future outcomes, it was trained on a 80/20 time based train test split to prevent data leakage. We engineered various features to capture finer details of team momentum and efficiency:

- Possessions (pace): $\text{fieldGoalsAttempted} + (0.44 \times \text{freeThrowsAttempted}) - \text{reboundsOffensive} + \text{turnovers}$. This normalizes the scoring against the pace of play (so fast paced teams are not falsely rewarded as the better team because they take more shots per game)
- Home Strength Rating: Expanding the mean of a team’s home games only, since Rest and Home/Away context are statistically big contributors to point margin, Entine & Small (2007)[12] found that the differential rest between home away teams is a big factor when it comes to NBA home court advantage.
- Fatigue Index: A penalty based on the days of rest and if they are playing at home or away. This is shown as teams perform worse on the second night of a back to back or if there is only a day of rest but they have to travel a long distance.
- 10-Game Rolling Averages: teamScore, opponentScore, fieldGoalsPercentage, possessions, reboundsTotal, and assists rolled over the prior 10 games using .shift(1) to prevent leakage. This feature is used to get the standard for how the team has been recently playing.
- Short-Window Form Features (L3 / L5): 3 game and 5 game rolling averages of team score, opponent score, and net margin, these get the recent momentum of the team as they go into the game, to better capture the team’s “energy” or “momentum” that can’t really be shown in the last 10 games.
- Opponent-Mirrored Statistics: All rolling and short-window features were mirrored for the opposing team. Ouyang et al. (2024)[11] talk about how important it is to look at the opponents’ features because the game outcome is found through “team strength” not performance in isolation.

The GBR was configured with 400 estimators, a learning rate of 0.03, max depth of 3, and subsample rate of 0.7 to reduce overfitting.

TESTING STRATEGY

Our testing strategy separated each test plan according to its corresponding functional area

Test Plan 1: UI & Dashboard (FA1)

Test: Verify Game Schedule: Validates date filtering and empty-state handling (no games).

Test: Verify Arbitrage Logic: Validates that "Edge" calculations are mathematically correct and negative edges are hidden.

Test: Verify Player Props: Validates the search filter and API market restrictions.

Test Plan 2: Predictive Modelling (FA2)

Test: Verify Data Ingestion: Validates that Team IDs are normalized and missing values are imputed.

Test: Verify Model Accuracy: Assert that the Random Forest Regressor achieves a Mean Absolute Error (MAE) < 12.0 points.

Test: Verify Daily Pipeline: Validates that the automated script successfully generates the JSON output file.

Test Plan 3: User Management (FA3)

Test: Verify Google Auth: Validates the OAuth redirection and user session persistence.

Test: Verify Discussion Threads: Validates that empty comments are rejected (Validation Logic) and posts are timestamped.

Test: Verify Favorites: Validates that adding a "Star" persists the preference to the Firestore database.

On top of the the above functional test plans, many backend tests were done, they check: live API routes and evaluate the Models against real data. The test_api_routes.py has 20 tests make sure all REST endpoints have the correct structure, fields and the correct error code for the historical data, live odds, prediction history, player prop, arbtrage results, google oauth and user profile updates, and they all pass the screenshot below:

```

Accuracy = 0.7731
petermina@Peters-MacBook-Pro Open-Bet % pytest backend/tests/test_api_routes.py -v
test session starts
platform darwin -- Python 3.12.5, pytest-9.0.3, pluggy-1.6.0 -- /Library/Frameworks/Python.framework/Versions/3.12/bin/python3.12
cachedir: .pytest_cache
rootdir: /Users/petermina/Desktop/Capstone/Open Bet/Open-Bet
plugins: anyio-4.6.0
collected 20 items

backend/tests/test_api_routes.py::TestHistoricalData::test_returns_list_without_date PASSED [ 5%]
backend/tests/test_api_routes.py::TestHistoricalData::test_each_game_has_required_fields PASSED [ 10%]
backend/tests/test_api_routes.py::TestHistoricalData::test_date_filter_returns_only_matching_games PASSED [ 15%]
backend/tests/test_api_routes.py::TestHistoricalData::test_unknown_date_returns_empty_list PASSED [ 20%]
backend/tests/test_api_routes.py::TestLiveNBAAdds::test_returns_list_of_games PASSED [ 25%]
backend/tests/test_api_routes.py::TestLiveNBAAdds::test_game_objects_have_expected_keys PASSED [ 30%]
backend/tests/test_api_routes.py::TestLiveNBAAdds::test_missing_file_returns_404 PASSED [ 35%]
backend/tests/test_api_routes.py::TestPredictionHistory::test_returns_dict_with_games_key PASSED [ 40%]
backend/tests/test_api_routes.py::TestPredictionHistory::test_games_list_is_non_empty PASSED [ 45%]
backend/tests/test_api_routes.py::TestPredictionHistory::test_prediction_record_has_required_fields PASSED [ 50%]
backend/tests/test_api_routes.py::TestPredictionHistory::test_is_correct_is_boolean PASSED [ 55%]
backend/tests/test_api_routes.py::TestPlayerProps::test_returns_list PASSED [ 60%]
backend/tests/test_api_routes.py::TestPlayerProps::test_prop_objects_have_player_field PASSED [ 65%]
backend/tests/test_api_routes.py::TestPlayerProps::test_prop_objects_have_market_field PASSED [ 70%]
backend/tests/test_api_routes.py::TestPlayerProps::test_edge_vs_line_is_numeric_when_present PASSED [ 75%]
backend/tests/test_api_routes.py::TestArbitrage::test_returns_list_when_arb_found PASSED [ 80%]
backend/tests/test_api_routes.py::TestArbitrage::test_arb_result_fields PASSED [ 85%]
backend/tests/test_api_routes.py::TestGoogleAuth::test_missing_token_returns_error PASSED [ 90%]
backend/tests/test_api_routes.py::TestGoogleAuth::test_missing_body_returns_error PASSED [ 95%]
backend/tests/test_api_routes.py::TestUserUpdate::test_missing_uid_returns_400 PASSED [100%]

===== 20 passed in 0.43s =====

```

Figure 5.1: test_api_routes.py output

For the API, there are 2 scripts to "evaluate" the player prop model and team model, these tests return the MAE, RMSE, R² for points, rebounds, assists, and thee pointers, also with confusion matrix metrics (TP, FP, TN, FN) going

against ground truth game outcomes which goes to a csv file. For the Team model there is an 80/20 split and historical winner rule, there are 53 tests for the Player prop model and 18 for the team model. They check all metrics are in normal NBA ranges and the confusion matrix is consistent with prediction history, the false positive rates match $FPR = FP/(FP+TN)$ (this is the formal definition) and again this is all going to a csv file. So in total there is 71 tests done and the outcomes (all passes) give us the needed false positive rates against ground truth data and the outcomes for both the player prop and team model.

```
petermina@Peters-MacBook-Pro Open-Bet % python3 -m backend.evaluation.evaluate_player_props
Rows loaded: 125 and Players: 125

points:
MAE=4.326 RMSE=5.855 R²=0.3764
Binary rule: pred > median (7.0) = OVER signal
TP=19 FP=1 TN=4 FN=1
FPR=0.2000 Precision=0.9500 Recall=0.9500

reboundsTotal:
MAE=3.395 RMSE=4.276 R²=0.0986
Binary rule: pred > median (3.0) = OVER signal
TP=11 FP=10 TN=3 FN=1
FPR=0.7692 Precision=0.5238 Recall=0.9167

assists:
MAE=2.369 RMSE=3.150 R²=0.2299
Binary rule: pred > median (1.0) = OVER signal
TP=18 FP=4 TN=3 FN=0
FPR=0.5714 Precision=0.8182 Recall=1.0000

threePointersMade:
MAE=1.380 RMSE=1.645 R²=0.9340
Binary rule: pred > median (0.5) = OVER signal
TP=12 FP=1 TN=2 FN=10
FPR=0.3333 Precision=0.9231 Recall=0.5455

Predictions saved → /Users/petermina/Desktop/Capstone/Open Bet/Open-Bet/backend/evaluation/results/player_props_predictions.csv
petermina@Peters-MacBook-Pro Open-Bet % python3 -m backend.evaluation.evaluate_team_model

OpenBet Team Model Evaluation
Training on 116,747 rows, testing on 29,187 rows...

Regression Metrics (held-out 20%):
MAE : 9.418 points
RMSE : 11.914 points
R² : 0.1915

Predictions saved to /Users/petermina/Desktop/Capstone/Open Bet/Open-Bet/backend/evaluation/results/team_predictions_vs_actuals.csv

Confusion Matrix (N=1139 predictions from history):
Binary rule : predicted_winner == home_team
TP=510 FP=204 TN=309 FN=116
FPR = 0.3977 (204/513)
Precision = 0.7143
Recall = 0.8147
Accuracy = 0.7191
```

Figure 5.2: Player Prop and Team Model Test outcomes

VALIDATION RESULTS

The OpenBet system has been verified against the functional use case requirements. Results confirm the basic potentially shippable product increment. Overall Stability

The systems main summary use case is met from the user's journey from loading the initial homepage to selecting a date and viewing the schedule games odds and prediction. The system demonstrates 100% uptime and availability. Zero crashes have been reported during testing.

Model Evaluation

The model was evaluated using a 80/20 time based train test split, the 20% is the test set which is the games that occurred after all training games, this way prevents data leakage and makes sure that the performance metrics show the real power on future games.

For the Beta release, the change was made to the game prediction model from Random Forest Regressor to a Gradient Boosting Regressor, also we tried to improve on the Player Prop model but there was no big improvement so nothing was changed about it.

The following tables show the performance of both versions of the Game Prediction Model:

Metric	Random Forest (Old Version)	Gradient Boosting (New Verion)
MAE	9.59 points	9.41 points
RMSE	12.13 points	11.91 points
R ²	0.163	0.192

Table 3.1: Overall Performance (all seasons)

Metric	Random Forest (Old Version)	Gradient Boosting (New Verion)
MAE	10.17 points	9.96 points
RMSE	12.90 points	12.70 points
R ²	0.178	0.207

Table 3.2 Current Season Performance

The Gradient Boosting model shows improvement across all testing for both the historical/overall dataset and also this current season (2025-2026). R² is improved from 0.163 to 0.192 so now there is 19% variance in final scores for games, this aligns with competitive baselines established in similar data science studies [9]. We also see MAE decreasing so we know that the Fatigue index, short window L3/L5 and opponent mirrored stats features all help predicting the outcome of games.

User Testing

Peer testing with fellow computer science students returned feedback for the easy functionality of the date picker and odds displays.

COST ESTIMATION & FEASIBILITY ANALYSIS

At the moment, OpenBet is completely based on free tier services, including Render as backend, The Odds API for odds, Google Cloud Firestore as database, and Google OAuth through Firebase authentication for sign-in. So currently there are no charges, but the free tiers are limited in use and would not be enough for a large website where multiple people visit it at once. In case one wants to commercialize it or make it bigger, we will go through a monthly cost estimate and how much to charge users to be able to afford it.

Render provides the Python/Flask backend. The free pricing tier sleeps the service if inactive and supplies just 512 MB RAM and 0.1 CPU, making it unsuitable for an always-on production environment. The starting paid tier starts at \$7 per month and guarantees the service to be always on. The standard pricing tier (which is better for production purposes) offers 2 GB RAM and 1 CPU and costs \$25 per month. For small to medium production deployment, this standard tier can be considered a minimum.

Odds API provides live bookmaker odds. Free tier limits the number of calls to 500 per month, which if we were to start calling the API more often for more accurate information (needed for more users) we would run out quite quickly. Production system that regularly updates odds for a live audience requires the 20K tier at \$30 per month. If odds have to be polled more often, the 100K tier at \$59 monthly would be needed, but the \$30 tier is good to start out with as we do manage to stretch the 500 for a few weeks currently.

Google Cloud Firestore acts as a database for storing discussion threads and user-related information. Free-tier Spark limits one to 50,000 reads and 20,000 writes per day, which may be enough for some cases. Under paid Blaze tier, one pays \$0.18 per 100,000 reads, \$0.18 per 100,000 writes, and \$0.26 per gigabyte storage. With a small production application and a few hundreds of MAUs reading threads and writing information, it may cost around \$5-\$10 monthly on top of free daily read and writes.

Authentication through Google / Firebase is the easiest pricing model. Under both free Spark tier and Blaze tiers, Google sign-ins are available for free to the first 50,000 MAUs. Also, usage of Google authentication is free up to that limit as a social login provider. If OpenBet were ever to go over the 50,000 monthly logged-in users, Google/Firebase would then start charging only for the additional MAUs above that 50k mark, at 0.55 cents per extra active user per month. So at 60,000 MAUs the first 50,000 would be free and only the extra 10,000 users then have an additional cost of about \$55 USD/month. We will estimate under 50,000 users per month as to go over we would probably need to upgrade other services, and we are just going for the first paid tiers of these services for this estimate.

Service	Free Tier limit	Recommended tier	Monthly Cost
Render (Web Service)	512 MB RAM (spins down on idle)	Standard: 2 GB RAM, 1 CPU (always-on)	\$25 per month
The Odds API	500 requests/month	20K credits/month	\$30 per month
Google Cloud Firestore	50K reads / 20K writes per day	Blaze tier (pay as you go, estimated light usage)	Around \$5-\$10 per month
Google OAuth (Firebase Auth)	50,000 MAU/month	Pay as you go (0.55 cents per user past 50,000 MAUs)	\$0
Total			\$60-65 per month (rough estimate)

Table 3.3: Monthly Running Cost

Per User Cost

The average cost of operation per month comes to \$60-\$65 per month. The cost per user is dependent on the number of customers. Below is a table that shows the lowest possible monthly cost per customer that OpenBet can charge its users to pay for the services:

Monthly Users	Monthly Cost	Cost Per User
100	65	\$0.65
500	65	\$0.13
1000	65	\$0.065
10,000	65	\$0.0065

Table 3.4: Per User Cost Breakdown

As seen from these numbers, the price per user is quite low because of this the freemium business model looks like a good plan, which means the core product is available for free, but access to premium features (historical predictions,

player prop model prediction, social aspects, or arbitrage opportunities), will cost somewhere between \$3.99 and \$4.99 monthly, covers the service cost easily (with a minimum of 20 paying users) Even though OpenBet is free, it has a low monthly operating cost which is good to keep the main core of the project available to people, which is to inform people and aggregate information. So to be able to upgrade the services and backend and even reach profitability at a low user count is a very good finding.

CONCLUSION

We set out to give transparency in the sports betting industry, to provide a clear aggregate of betting lines and to inform users, as well as informing them of “bad bets”. We have completed the first part and people can see the days' lines and can access the model, but there is much more to be done. The model itself must be much more complex to be able to rival the sportsbooks and the arbitrage, and player props pages need much more work as currently they aren't doing their jobs.

PROJECT SUITABILITY

We believe OpenBet to be highly suitable and feasible solution for the current market.

Technical Feasibility

As of now the project is built entirely on zero cost technologies. Free tiers of Render hosting services, Google Firestore database services and The Odds-API the project has \$0 in operating costs. We believe this proves it can exist and provide value in the real world. It can serve as a free educational research for any NBA bettor. The data pipeline aims to be completely autonomous to keep odds and predictions accurate and up to date.

Domain Feasibility

Our current resources provide us insight into the competitive basketball statistical domain via our expert Samuel Palubisili. He believes our current model demonstrates a strong mathematical prediction of NBA basketball matches.

Stakeholder Impact

We believe our project demonstrates the successful application of Advanced machine learning, Data analytics and Cloud computing practices to solve a real world data science problem

DOMAIN EXPERT EVALUATION

Over the development of OpenBet we were lucky to have been supported by experts in both the sports analytics field and Web development field. Sam P was our first official expert to review our work and provide insight. He was able to provide us with current industry practices with regards to basketball statistics. He was imperative for the development of the features used in our model. He helped guide us in creating features relative to our current expertise in machine learning. He also gave us insight into the products and services used by professional teams at the collegiate level as well as Team Canada basketball. Sam was an immense help in setting up the model for this release and will no doubt play a huge role in the complexity additions in the future. Our second expert we found later in the development cycle but he still played a major role in the development of OpenBet. Simon McNeil brought much needed experience and insight to the development process of OpenBet. While initially impressed by our idea and current progress we made he instantly saw areas for improvement. The biggest change he brought about was the batch model update approach to solve our API dependency problem. This immediately removed a roadblock for us making the rest of the project elaboration release much easier. He was also impressed with our current iteration

of our Random forest regression model. Simon will continue to play a huge part in the future of OpenBets development.

USER TESTIMONIALS

Our current user feedback work has all been gathered in person. Our sample has strictly been Sheridan peers and capstone students. Feedback across all testers have found the display to be simple yet informative. The biggest request along the UI front has been for more reactive web viewing and improved mobile viewing. Testers appreciated the odds displays with many remarking how it follows the popular card format various gambling platforms use. Further surveys and tests must be conducted regarding model performance/accuracy sentiments

ALPHA RELEASE

For the Alpha Release the following has been added:

- Users and User management
- Integrating with Firestone
- Discussion boards
- Favorite Teams
- Google Login
- Player Props Model

The following has been updated:

- Player Prop Page to display the odds correctly
- Arbitrage page to show real arbitrage opportunities

Random Forest was the modeling technique used for the player props prediction (similarly to the Team Prediction). This is because ensemble methods have strong generalization performance but remaining robust to overfitting on complex classification issues. Breiman (2001)[13] is where random forest is introduced and shows that when aggregating many random decision trees, the variance is reduced and get much lower error rates (it is also resistant to noise from the data) Other studies into the Random Forest algorithm say that this ensemble structure gives an accurate prediction with low bias and low variance on tabular data, which makes it better for data sets used in sports analytics because it's so feature heavy/dense (Korn, 2024)[14] In the sports world random forests have shown strong performance when it comes to predicting basketball game outcomes and team win rates, which how good they are for feature selection and model optimization for basketball analytics (Li et al. 2025)[15]. This is what we want and working towards to have a model that can predict player specific performances based on odds set by sportsbooks.

BETA RELEASE

For the Beta Release the following has been added:

- The following has been updated:

- Visual Language of each page so there is no UI inconsistencies
- Automation of the Project so it can update on its own
- Updated the Game Prediction Model

In the Beta Release we worked on three major things: Updating/Syncing the UI, Automating the project, and researching and the subsequent upgrading of the model(s). The visual language of the site was not the same and it was jarring to go from one page to the next, so the UI was unified so each page has a title and subtitle that all match, as well as updating the header to better accommodate the Google Login. There were actual changes made to the Profile Page as that was very barebones in the alpha release, so it was cleaned up and also matches the visual language of the rest of the project. The Game Prediction Model was also updated, with a change from Random Forest Regressor to Gradient Boosting Regressor, and incorporating some new features in the model which gave a strong improvement to the predictive accuracy. Lastly the site is now completely automated and can run on its own, with every morning getting the previous day's outcomes using the Kaggle API, then the OddsAPI is called so the day's games and odds can be seen, finally the API data is run through the models to give "our predictions" and pushing this all live to the frontend. This is all done through the Github Actions Workflow and runs every morning, this makes OpenBet fully sustainable and able to run on its own with no work needed.

APPENDICES

Software Requirements Specification (SRS): [Link](#)

Software Design Document (SDD): [Link](#)

Software Test Plan (STP): [Link](#)

BIBLIOGRAPHY

- [1] Grand View Research. (2025). Sports Betting Market Size, Share & Trends Analysis Report, 2030. Link: <https://www.grandviewresearch.com/industry-analysis/sports-betting-market-report>
- [2] Investopedia. (2025). Arbitrage: Definition, Examples, and How Investors Use It. Link: <https://www.investopedia.com/terms/a/arbitrage.asp>
- [3] Statistics Canada. (2022). North American Industry Classification System (NAICS) Canada 2022 - 7132 Gambling Industries. Link: <https://www23.statcan.gc.ca/imdb/p3VD.pl?Function=getVD&TVD=1381557&CVD=1381568&CPV=7132>
- [4] Data Bridge Market Research. (2024). Canada Sports Betting Market Size, Trends and Forecast to 2032. Link: <https://www.databridgemarketresearch.com/reports/canada-sports-betting-market>
- [5] iGaming Ontario. (2024). FY 2023-24 Full Year Market Performance Report. Link: <https://igamingontario.ca/en/news/igaming-ontarios-fy-2023-24-full-year-market-performance-report>
- [6] Legal Sports Report. (2025). *US Sports Betting Revenue & Handle Tracker: 2024 Record Breaking Year*. Link: <https://www.legalsportsreport.com/sports-betting/revenue/>
- [7] Canadian Gaming Business. (2026). *AGCO fines FanDuel \$350K for failing to detect suspicious betting activity*. Link: <https://www.canadiangamingbusiness.com/2026/01/08/agco-fanduel-fine-bets>
- [8] University of Rochester. (2024). *Basketball Player Performance Prediction Using Random Forest*. Link: https://hajim.rochester.edu/ece/sites/zduan/teaching/ece408/projects/2024/KevinFobare_MukhtarSuleman_RyanPappania_RodneyBoone_FinalProject_Report.pdf
- [9] Towards Data Science. (2025). *Using Random Forest Regression to predict NBA Players scoring output*. Link: <https://medium.com/inst414-data-science-tech/using-random-forest-regression-to-predict-nba-players-scoring-output-6bf7995e169e>
- [10] <https://www.kaggle.com/datasets/eoinamoore/historical-nba-data-and-player-box-scores/data?select=Games.csv>
- [11] Ouyang Y, Li X, Zhou W, Hong W, Zheng W, Qi F, et al. (2024) Integration of machine learning XGBoost and SHAP models for NBA game outcome prediction and quantitative analysis methodology. PLoS ONE 19(7): e0307478. <https://doi.org/10.1371/journal.pone.0307478>
- [12] Entine, O., & Small, D. (2008). The role of rest in the NBA home-court advantage. Journal of Quantitative Analysis in Sports, 4(2). <https://faculty.wharton.upenn.edu/wp-content/uploads/2012/04/Nba.pdf>
- [13] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5-32. doi:<https://doi.org/10.1023/A:1010933404324>
- [14] Korn, J. (2020). Ensemble classification: An analysis of the random forest model (Honors research project). The University of Akron
- [15] Li, C., Zhang, H., Zhang, Y., Shen, J., & An, R. (2025). The application of artificial intelligence techniques in predicting game outcomes of professional basketball league: A systematic review. PLoS One, 20(6), 28. doi:<https://doi.org/10.1371/journal.pone.0326326>